

SOSUNOVO, Russian FederationWMO: **318660**Lat: **46.5456N**Lon: **138.3381E**Elev: **4**StdP: **101.27**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-20.7	-31.4	0.2	-19.6	-29.9	0.2	-18.3	16.8	-6.0	13.8	-9.1	3.4	260	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	24.8	19.1	22.5	18.0	20.5	17.1	20.0	23.1	19.0	21.2	18.0	19.9	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.8	21.1	18.1	13.0	20.1	17.2	12.3	19.1	57.6	23.1	54.2	21.3	51.2	20.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.0	8.6	DB	-27.1	29.6	1.8	1.8	-28.5	30.9	-29.5	32.0	-30.6	33.0	-31.9	34.3	(4)
(5)			WB	-27.4	22.4	1.7	0.5	-28.6	22.8	-29.6	23.1	-30.6	23.4	-31.9	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-11.9	-9.9	-4.1	2.0	6.6	10.5	15.4	17.4	13.5	6.2	-2.6	-10.1	(6)	
	DBStd	10.63	4.25	4.44	3.82	2.89	3.41	2.71	2.70	2.19	2.90	3.28	5.37	4.43	(7)	
	HDD10.0	3186	680	556	437	242	117	23	0	0	6	123	378	624	(8)	
	HDD18.3	5686	938	789	696	491	362	234	98	44	146	376	628	883	(9)	
	CDD10.0	569	0	0	0	1	13	39	168	230	112	6	0	0	(10)	
	CDD18.3	26	0	0	0	0	0	1	8	16	2	0	0	0	(11)	
	CDH23.3	135	0	0	0	0	3	3	65	58	6	0	0	0	(12)	
	CDH26.7	22	0	0	0	0	1	0	17	5	0	0	0	0	(13)	
(14) Wind	WSAvg	4.1	4.8	4.4	4.2	4.0	4.1	3.7	3.3	3.3	3.8	4.1	4.4	5.0	(14)	
(15) Precipitation	PrecAvg	749	16	10	29	44	90	72	108	124	107	77	44	27	(15)	
	PrecMax	1391	61	33	98	244	154	464	326	523	296	242	230	89	(16)	
	PrecMin	252	0	0	5	4	5	8	12	16	15	4	1	0	(17)	
	PrecStd	327	15	7	24	45	39	85	78	124	84	56	60	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	3.6	9.4	17.1	21.4	22.5	29.1	27.2	24.3	18.6	12.0	3.2	(19)	
		MCWB	-1.1	-1.3	3.5	8.8	12.0	17.5	20.7	20.0	17.9	11.1	5.6	1.3	(20)	
	2%	DB	-2.0	0.4	5.6	12.1	17.9	18.9	24.8	25.0	21.7	15.6	8.5	0.1	(21)	
		MCWB	-3.8	-2.6	0.7	5.9	10.3	14.9	19.0	19.6	16.2	10.2	4.1	-1.9	(22)	
	5%	DB	-3.7	-1.2	3.0	9.2	14.9	16.5	21.3	22.9	19.8	13.7	6.6	-1.6	(23)	
		MCWB	-5.8	-3.3	-0.5	4.0	9.2	14.0	17.5	19.1	15.5	9.4	3.9	-3.5	(24)	
	10%	DB	-5.3	-2.7	1.3	7.0	11.9	14.4	18.7	21.1	18.5	12.3	5.1	-3.2	(25)	
		MCWB	-7.1	-4.8	-1.2	2.7	8.1	12.2	16.9	18.2	15.2	8.9	2.8	-5.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	-0.5	3.9	8.7	12.9	17.9	21.5	21.5	18.9	13.2	6.9	1.2	(27)	
		MCDB	-0.4	1.6	9.2	15.8	19.0	21.9	27.5	24.3	22.3	15.4	8.7	2.3	(28)	
	2%	WB	-3.8	-1.9	1.3	6.3	11.0	15.9	19.4	20.5	17.7	11.7	5.5	-1.4	(29)	
		MCDB	-2.3	0.0	4.4	11.3	16.3	18.0	23.8	23.1	20.0	14.0	7.1	-0.5	(30)	
	5%	WB	-5.5	-3.2	0.1	4.5	9.5	13.9	18.0	19.6	16.8	10.8	4.4	-3.3	(31)	
		MCDB	-4.0	-1.2	2.0	8.1	14.3	16.0	20.0	21.7	18.8	12.7	6.3	-1.9	(32)	
	10%	WB	-7.0	-4.7	-0.8	3.3	8.2	12.4	17.1	18.8	15.8	9.6	2.8	-5.0	(33)	
		MCDB	-5.2	-2.6	1.0	6.2	11.2	14.1	18.6	20.6	17.8	11.9	4.9	-3.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.9	8.5	7.1	6.6	4.9	4.6	5.6	8.7	10.0	8.8	8.7	(35)	
		MCDBR	9.4	10.6	10.1	12.1	12.9	10.0	11.3	9.9	10.6	12.8	10.4	9.4	(36)	
	5% WB	MCWBR	8.1	8.4	6.6	7.0	6.6	6.4	6.2	5.3	6.2	8.2	8.1	8.0	(37)	
		MCWBR	8.8	9.5	8.5	11.0	11.6	9.6	8.3	7.5	7.9	9.1	9.2	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.306	0.356	0.428	0.515	0.546	0.524	0.484	0.452	0.406	0.404	0.358	0.320	(40)	
		taud	2.204	2.052	1.915	1.807	1.828	1.988	2.169	2.270	2.313	2.148	2.174	2.159	(41)	
	Ebn at Noon	Ebn at Noon	764	783	774	742	737	758	786	797	802	723	684	687	(42)	
		Edh at Noon	84	122	163	200	202	174	143	125	109	110	88	80	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.66	2.68	3.90	4.60	4.77	4.83	4.51	4.44	4.01	2.81	1.81	1.36	(44)	
	RadStd	RadStd	0.14	0.18	0.35	0.53	0.48	0.69	0.47	0.39	0.37	0.23	0.16	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page